



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Problem Sheet I

AY 25-26

Semester II

1. Let $G = \{0, 1, 2, 3\}$ with the binary operation $*$ defined by

$$a * b = (a + b) \bmod 4.$$

- (a) Verify whether $(G, *)$ is a group.
 - (b) Find the identity element and inverse of each element.
2. Find the group of units modulo 10, denoted by $U(10)$.
- (a) List all elements of $U(10)$.
 - (b) Find the inverse of each element.
 - (c) Verify that $U(10)$ forms a group under multiplication modulo 10.
3. Show that the set of rational numbers $\mathbb{Q}$ forms a field under usual addition and multiplication.
4. Verify whether $\mathbb{Z}_7$ is a field. Find the multiplicative inverse of 3 in $\mathbb{Z}_7$.
5. Check whether $\mathbb{Z}_6$ is a field. Justify your answer.
6. Check whether V is a vector space or not for each of the cases. *True or False.*
- (i) $V = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = 0\}$, $F = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}$.
 - (ii) $V = \{x = (x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}$, $F = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
 - (iii) $V = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 = 0\}$, $F = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
 - (iv) $V = \{x \in \mathbb{R} : x \geq 0\}$, the set of non-negative real numbers, and $F = \mathbb{R}$. For $x, y \in V$, $\alpha \in \mathbb{R}$, $x + y := xy$, $\alpha x := |\alpha|x$.
 - (v) $V = \mathcal{R}([a, b], \mathbb{R})$, the set of all real-valued Riemann integrable functions on $[a, b]$, and $F = \mathbb{R}$, with usual addition and scalar multiplication of functions.
 - (vi) V is the set of all polynomials of degree 5 with real coefficients and $F = \mathbb{R}$, with usual addition and scalar multiplication of polynomials.
7. In the following, vector space V over $F = \mathbb{R}$ is given with usual operations on it. Check whether the given subset U is a subspace of V .
- (a) $V = \mathcal{P}_3$ and $U = \{a_0 + a_1 t + a_2 t^2 + a_3 t^3 : a_0 = 0\}$. Where $\mathcal{P}_3$ denotes the set of all polynomials of degree 3.
 - (b) $V = \mathcal{P}_3$ and $U = \{a_0 + a_1 t + a_2 t^2 + a_3 t^3 : a_2 = 0\}$.
 - (c) $V = \mathbb{R}^3$. Let U be the all linear combinations of $\mathbf{v} = (1, 4, 0)$ and $\mathbf{w} = (2, 2, 2)$.